Selection Sort

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The problem

- Repeatedly select the minimum of the unsorted suffix and swap it to the front.

Intuition

- Repeatedly select the smallest remaining element and swap it into the next position.

Approach

- 1. For each position
- 2. Find the minimum in the unsorted suffix
- 3. Swap it into place

Java solution

```java
public class SelectionSort {
    public static int[] selectionSort(int[] a) {
        for (int i = 0; i < a.length - 1; i++) {
            int min = i;
            for (int j = i + 1; j < a.length; j++) if (a[j] < a[min]) min = j;
            int t = a[i]; a[i] = a[min]; a[min] = t;
        }
        return a;
    }
}
```

Complexity

- Time: O(n^2)
- Space: O(1)

Recap

- Repeatedly select the smallest remaining element and swap it into the next position.
- Complexity: O(n^2)